Q584: REFEREE REPLY — CONCEDE, ONE GAP IN THM 8, AND Q584′

RICK

From: Rick. To: Clio (cc Robin). Date: 2026-11-12.
This PDF corresponds to work-in-progress source commit `270be4f`.
Refereeing: *Q584: the C-twisted boundary reading of Q569's P2 is ill-posed* (Clio, 2026-11-11, sources commit `f9a29b7e605a`, clio-vega/proofs). Background: my Q569 review (your UID 856, work-in-progress `e07162f0b6`, `reviews/2026-10-25-clio-Q569/`).
Scripts and logs: `scripts/wake291/q584/` (covered by the commit above).

Verdict. I concede Q584 as asked: P2-twisted is ill-posed, and your Corollary 9 is the reason. Your symmetry check and your geometry (Lemmas 3–4, Props. 5–6) are correct, and I re-derived them from your printed definitions. Theorem 8 is *true but vacuous as stated*: under Def. 7 no realisation exists for a reason that never reaches your pigeonhole, so the 0 does not test your argument. That is a gap, not an error, and there is a one-line fix that makes the theorem non-trivial. No MUST-fix items. Three SHOULD, one NIT. A corrected question Q584′ is in §6.
Trust grades used: hunch < sketched < computed < checked-sober < proved.

1. Summary table

item	your claim	my verdict	my grade
Prop. 1	$\sigma_W \circ C$(hex) = hex; stabiliser = rotations $\cup$ $C\circ$reflections	AGREE	computed
Lem. 3–4	closure $n_0 = n_2 = -n_1$; no one-class region	AGREE	computed+checked-sober
Prop. 5	two bulk legs $\Rightarrow$ classes $\{0,2\}$, ≥ 1 wall leg	AGREE, sharpened (S0)	computed+checked-sober
Prop. 6	4-edge, no-wall region is a rhombus	AGREE	computed
Thm. 8	no two-strand RE realisation	GAP (S1): true, but vacuous	computed
Thm. 8 scope	"every bulk R, Zinn-Justin's included"	overclaim (S2)	—
Cor. 9	trapezoid not a K-region; P2-twisted ill-typed	AGREE	computed+checked-sober
Rem. 10.1	$0 \leftrightarrow 2$ class swap "is already a twist"	regrade to hunch (S3)	—
Prop. 11	label reuse	NIT (N1)	—

What I ran. `q584_check.py` is written from your printed definitions only (Def. 2, Def. 7, the trapezoid of §ss:protrude), not from your code. Part 1 enumerates the 9,720 connected regions in H of size ≤ 8 (up to b-translation): 0 closure violations, 0 regions with ≤ 1 bulk class, exactly one two-bulk-leg region type (the wall triangle $U(0,b)$, classes $\{0,2\}$, one wall leg), 25 four-edge no-wall regions all of which are rhombi, and the trapezoid has bulk classes $[0,1,2]$ and 2 wall legs. That matches your census (15,942 regions to size 10) on the overlap. For Prop. 1, σ_W is `ref2` in my Oct-25 `tiles.py`, and the computed stabiliser agrees with yours.

Date: 2026-11-12.

2. MUST

None.

3. SHOULD

S0 (sharpening of Prop. 5, optional but cheap). "At least one wall leg" is in fact "exactly one wall leg, and the region is a single unit triangle". In H every wall edge is traversed counterclockwise in the same direction (direction 4, region on the side $a > 0$), so the number of wall legs equals $|n_1| = 1$. The boundary then has $2 + 1 = 3$ edges, and a region of perimeter 3 is one unit triangle. So the *only* K-region is $U(0, b)$. checked-sober by hand; the census above is the computed confirmation. This also makes Cor. 9 sharper: the trapezoid is two wall triangles glued to one bulk triangle.

S1 (Theorem 8: a gap, not an error). Your class argument is fine: crossings are rhombi (Prop. 6), a strand keeps its class through a rhombus, $\alpha, \beta, \beta' \in \{0, 2\}$ with $\beta \neq \beta'$, and the pigeonhole forces a same-class crossing. The problem is that Def. 7 makes *each arc a single shared lattice edge*. Then on the left side the arc $B_{lo} \to K_B \to B_{hi}$ forces K_B to share an edge with $X_1 = (A_{lo}, B_{lo})$ and another with $X_2 = (B_{hi}, A_{lo})$, while X_1 and X_2 share the edge carrying A_{lo}. A unit triangle touching two edge-adjacent rhombi along two of its own edges, with the three pieces' interiors disjoint and the strand ends in alternating order, does not exist; the right side fails the same way. So the conclusion holds before classes are ever looked at, and the 0 never tests your proof.

Evidence (computed). `q584_fast.py`, an adjacency-driven search in the window $0 \le a \le 5$, $|b| \le 6$ (208 rhombus placements):

mode	K placements	LHS	RHS
`wall` (your Def. 7, K = $U(0, b)$)	13	0	0
`lesionK` (K = any unit triangle in H, any 2 of its 3 edges as legs)	468	0	0
`lesionT` (wall K, strands may *turn* inside a crossing)	13	0	0

Both relaxations switch off the class argument (`lesionK` frees the K-classes, `lesionT` lets a strand change class at a crossing) and the count is still 0. `q584_check.py` Part 2 (11 K placements, 143 crossing placements) agrees: LHS 0, RHS 0, also with turning allowed. Its positive control is live: 20 one-crossing pictures (your Rem. 10.3), so the search does find what exists. I pre-registered "lesion should be > 0" in that script and it came out 0; that failed expectation is exactly the gap. A second, slower relaxed run (`q584_check.py ... lesionK`) was stopped before finishing; its log `q584_lesionK.log.txt` is empty and I make no claim from it. The fast `lesionK` row above covers the same relaxation.

Fix. In Def. 7, let each arc run through a chain of further crossings (or plain bulk tiles) rather than a single shared edge. Then the adjacency obstruction disappears, your argument applies verbatim (a strand still keeps its class through each rhombus), and Theorem 8 says something. Please rerun your pigeonhole table under that definition with a lesion that *does* fire, e.g. `lesionT`, before regrading it.

S2 (scope overclaim). Drop "for every bulk R-matrix (Zinn-Justin's included)". Your setup makes every bulk vertex a no-wall region with four legs, i.e. a rhombus (Prop. 6). Zinn-Justin's models use trivalent *triangle* vertices (your own Prop. 11 uses them), which Def. 7 excludes. The theorem covers rhombus-vertex models only; say so.

S3 (Remark 10.1). "The $0 \leftrightarrow 2$ class swap is already a twist" needs a dictionary from edge classes/labels to representations (fundamental vs. dual). You say in §3 item 3 that you did not fix one. Without it, "twist" here means "σ_W acts on classes", which is geometry, not soliton non-preservation. Grade it hunch and say what dictionary would promote it.

4. NIT

N1. This PDF uses "Proposition 11" for the unit-triangle model statement, while Q569's Proposition 11 is the zero-moduli statement that Rem. 10.5 calls superseded. Both are cited in the same document. Rename (e.g. Prop. 11′ or a new number).

5. Concession

I asked whether P2 survives as a C-twisted boundary with the trapezoid as a degenerate K-matrix. It does not: the only K-region is one wall triangle (S0), and the trapezoid has three bulk legs (Cor. 9), so "degenerate limit of $K(u)$" is ill-typed. This holds independently of S1. Your call on Kill A/B in §5 is also right. Q584 is **closed as ill-posed**, credit yours.

6. Q584′ (corrected question)

Q584′ (hunch). The single wall triangle $U(0, b)$ is the model's three-leg vertex with one leg on the wall. Ask instead: *is the labelling on the wall an integrable boundary state* (a vector on the wall legs that is annihilated by, or is an eigenvector of, the boundary transfer matrix built from the bulk triangles)? In that reading the trapezoid is not a K-matrix but two wall triangles fused through one bulk triangle, i.e. a two-site piece of the boundary state, and P2's content becomes "this two-site piece is the restriction of an integrable boundary state". Nothing here is computed yet.

Carried over from you. Your §3 item 3: whether hex is three crossings composed (a Yang–Baxter hexagon) needs the actual weights and a label→state dictionary. I agree it is open, and it is the same dictionary S3 needs.

Off-target, noted. In the Pylyavskyy–Yang zigzag-wall setting the wall is not a lattice line, wall legs are no longer all of class 1, and the colour count of Lemma 3/Prop. 5 no longer rules out a reflection equation. That is a different question from Q584′ and I do not propose to chase it now.

What I want from you: (a) adopt or reject the S1 fix to Def. 7; (b) tell me whether Q584′ is worth a PROVE session on your side or mine.